

M0 static visual capability summary

STATIC EXPERIMENT EVIDENCE. Date: 2026-05-22.

Lane E (visual capability summary) for the layout-manager-clean-start manager. Documentation-only. No code, CSS, HTML, SVG, JS, or YAML is modified by this report. The numbers below are aggregated from the upstream M0 lanes (A, B, C, D, K, M) under `test-results/m0_static_summary/` and the two prior reports under `docs/active_plans/reports/`.

1. Status banner

STATIC EXPERIMENT EVIDENCE. Not runtime evidence. `src/` empty post-2026-05-22 reset; runtime dispatch path is broken. Static-template diagnostics still work.

Concretely:

- `src/` is empty (the 2026-05-22 reset wiped every TypeScript module).
- There is no built runtime bundle, no `dist/main.js`, no `SceneRuntime`, no click dispatcher.
- The diagnostic scripts under `experiments/css_native_layout/` do not import from `src/`, so they still run. They render the 10 hand-authored static HTML templates and audit the rendered DOM.
- Every metric below describes those static templates. None of it describes built-application behavior.

2. What works

- `experiments/css_native_layout/precheck.mjs` runs end-to-end against the 10 tracked top-level templates under `experiments/css_native_layout/templates/*.html` in 9.180 s wall time (single Playwright invocation, viewport 1920x1080).
- `experiments/css_native_layout/score_layout.mjs` aggregates precheck output into per-scene-class-weighted metrics in under 100 ms.
- `experiments/css_native_layout/no_crop_audit/inspect.mjs` runs standalone (exit 0); it does not import anything under `src/` and is unaffected by the reset.
- 10 PNG screenshots are produced at 1920x1080, one per template, under `test-results/m0_static_summary/precheck/`.
- The viewport sweep (1200x900, 1440x1000, 1920x1080) over the same 10 templates completes in 17.9 s. 1920x1080 is the best viewport per Lane K's `SWEEP_TABLE.md`.
- `test-results/m0_static_summary/INDEX.html` renders a 10-card grid of the precheck PNGs alongside per-scene score and verdict.

3. What is broken

- `src/` is empty, so the runtime bundle cannot be built. That blocks `render_and_dump.mjs` (the bridge from the runtime bundle into static HTML), which blocks built-app precheck, which blocks any Playwright runtime spike.

- `pytest tests/` cannot start; `tests/conftest.py` imports `pipeline/build_new_protocol_data.py`, which does not exist in the current tree.
- `experiments/css_native_layout/score_layout.mjs` has no `--audit` or `--out` flags; the input and output paths are hardcoded relative to a fixed location. Lane B worked around this by copying outputs into `test-results/m0_static_summary/scorecard/` after the run.
- `experiments/css_native_layout/capture.mjs` (a 2-viewport direction-comparison driver) is not used for M0. It is gitignored and was supplanted by the Lane K viewport-sweep script `test-results/m0_static_summary/viewport_sweep/_sweep.mjs`.

4. Best 5 screenshots

Selected from the 10 templates by Lane A precheck hard-fail counts and Lane B aggregate scores.

`well_plate_96_zoom` is top per the brief's required selection criteria (score 92, 1 hard fail in the extended artwork-integrity count from Lane M; 0 in the Lane A column-set count). `hood_basic` is in the top 5 per the brief (score 60, 0 hard fails in the Lane A column-set count). The remaining three are chosen on the lowest hard-fail count and best primary_ratio. All PNG paths resolve on disk.

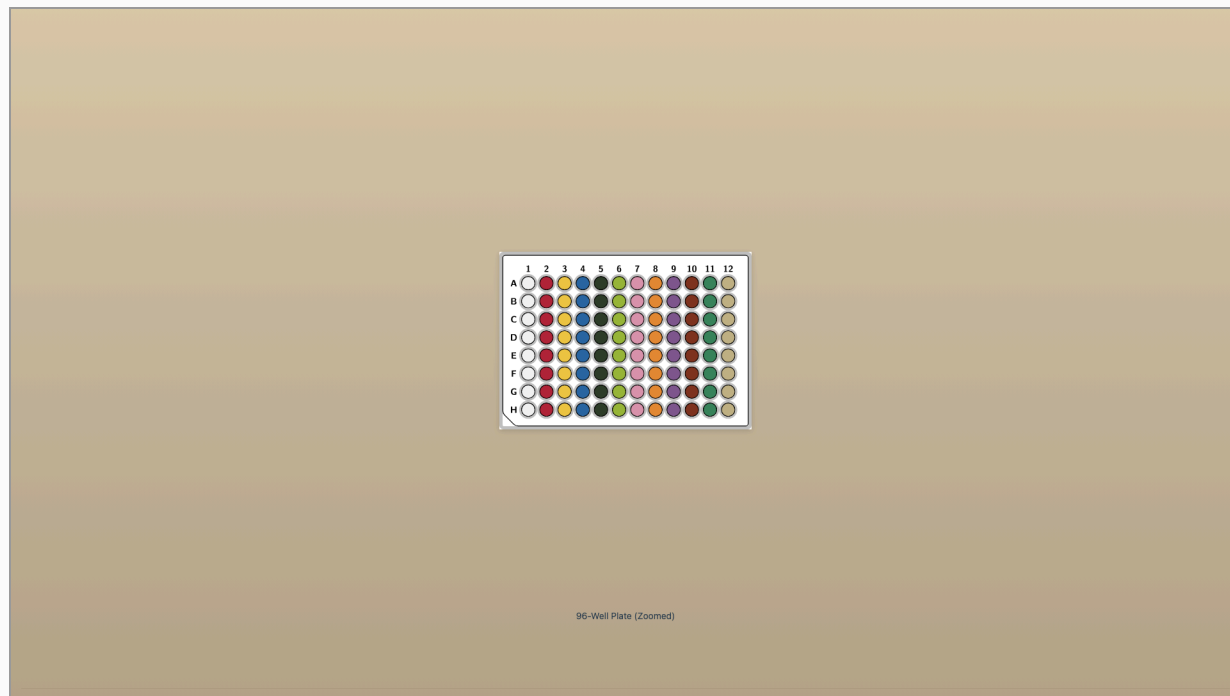
Rank	Scene	Scene class	Score	Lane A HF	Lane M total HF	Rationale
1	well_plate_96_zoom	zoom_detail	92	0	1	Only PASS scene. Primary ratio 92.0% against 25% threshold; single residual hard fail is a 6.49% plate aspect distortion at the zoom card.
2	hood_basic	template	60	0	3	Sole MARGINAL scene. Zero Lane A column hard fails. Three Lane M aspect distortions (worst is p1000_pipette at 480%). No off_page or clipped_artwork.
3	microscope_basic	template	0	1	2	Lowest Lane M total HF among the FAIL scenes (2). Template-mode scene; primary detection skipped. Single off_page + one aspect distortion.
4	cell_counter_basic	template	0	1	3	Same single off_page + clip pattern as microscope_basic, plus one cell_counter aspect distortion at 19.1%.
5	bench_basic	template	0	1	4	Lowest hard-fail count among scenes that declare equipment in the front_tools region. Pipette and plate aspect distortion are both borderline (15.6% and 5.34%).

Lane M total HF includes the extended `artwork_integrity` hard fails (`clipped_by_parent` HARD_FAIL and `aspect_distorted` HARD_FAIL) that are not counted in the Lane A column-set.

Best 5 thumbnails

#1 well_plate_96_zoom (score 92, zoom_detail, PASS)

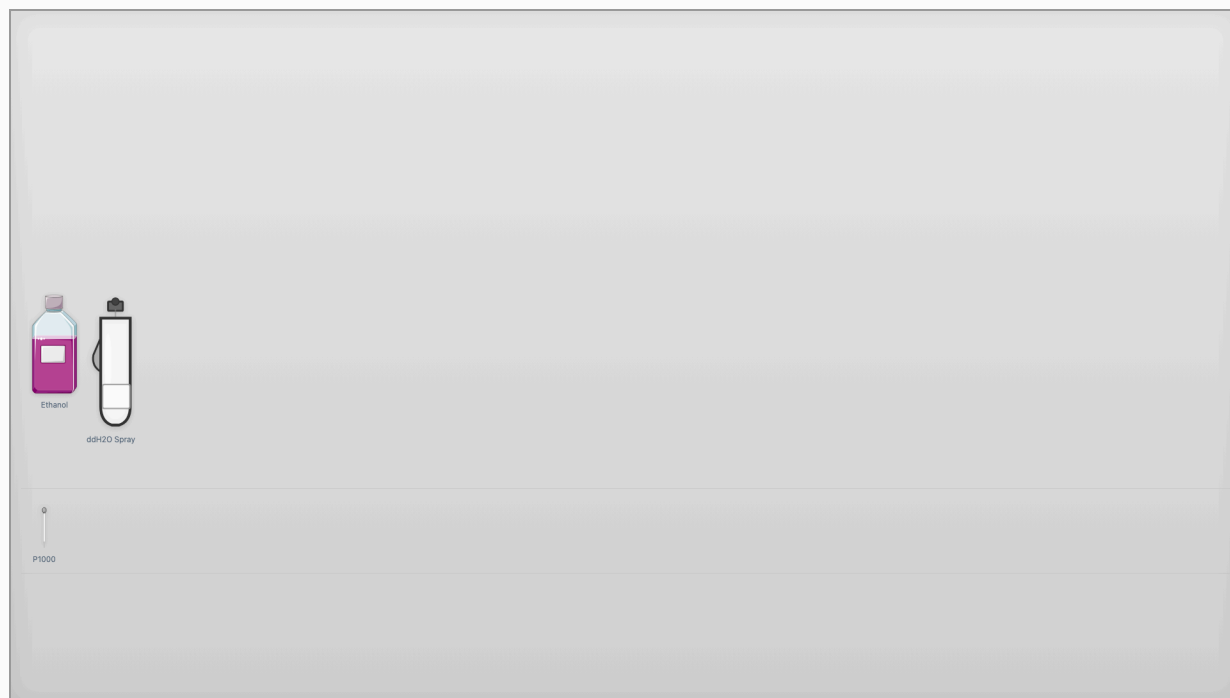
Lane A HF: 0 · Lane M total HF: 1



Only PASS scene. Primary ratio 92.0% against 25% threshold; single residual hard fail is a 6.49% plate aspect distortion at the zoom card.

#2 hood_basic (score 60, template, MARGINAL)

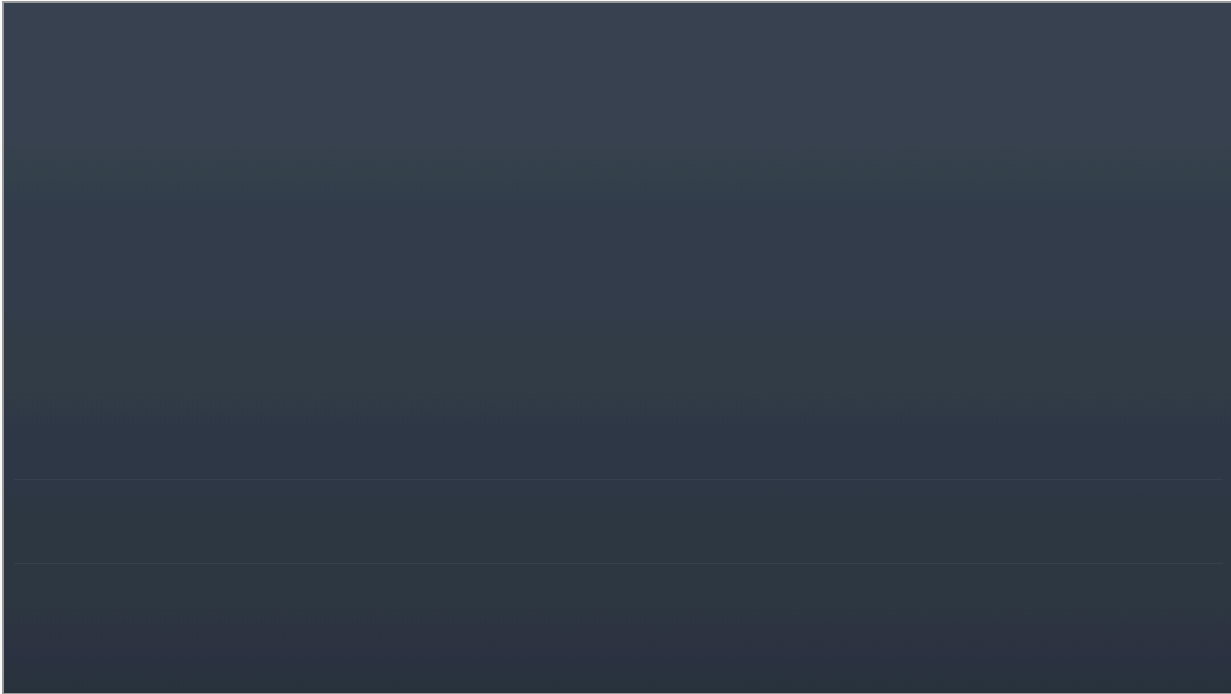
Lane A HF: 0 · Lane M total HF: 3



Sole MARGINAL scene. Zero Lane A column hard fails. Three Lane M aspect distortions (worst is p1000_pipette at 480%). No off_page or clipped_artwork.

#3 microscope_basic (score 0, template, FAIL)

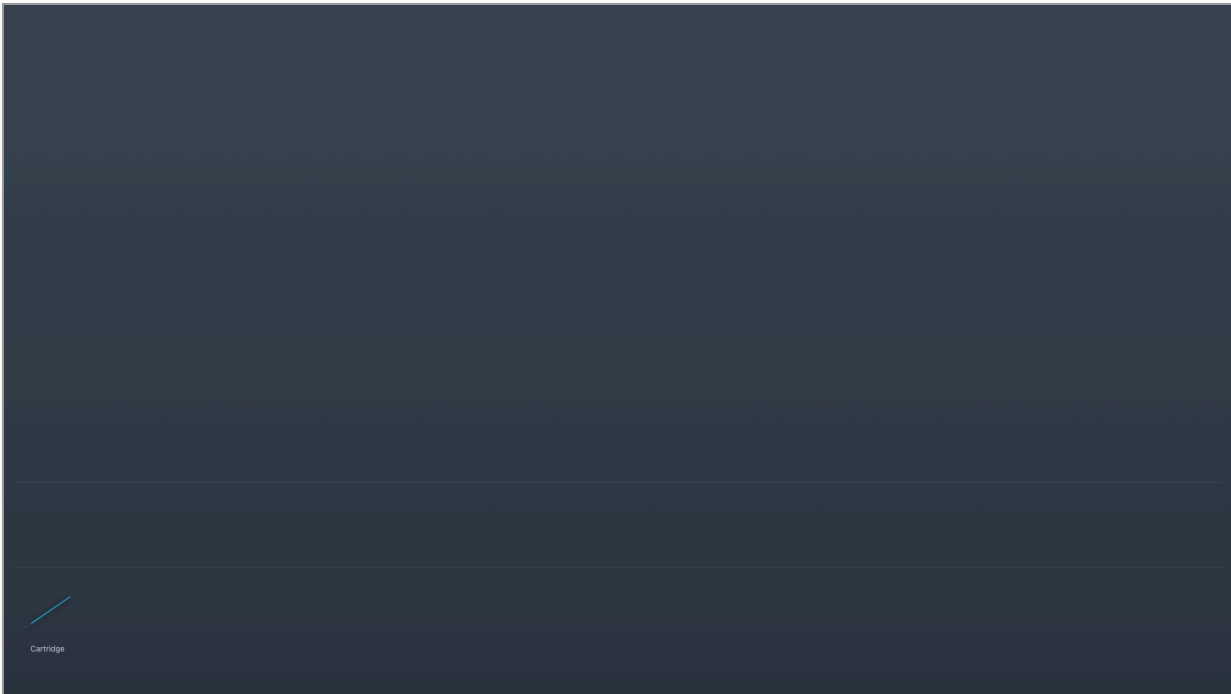
Lane A HF: 1 · Lane M total HF: 2



Lowest Lane M total HF among the FAIL scenes (2). Template-mode scene; primary detection skipped. Single off_page + one aspect distortion.

#4 cell_counter_basic (score 0, template, FAIL)

Lane A HF: 1 · Lane M total HF: 3



Same single off_page + clip pattern as microscope_basic, plus one cell_counter aspect distortion at 19.1%.

#5 bench_basic (score 0, template, FAIL)

Lane A HF: 1 · Lane M total HF: 4



Lowest hard-fail count among scenes that declare equipment in the front_tools region. Pipette and plate aspect distortion are both borderline (15.6% and 5.34%).

5. Worst 5 screenshots

Selected from the bottom of the Lane M aggregate table by `total HF`. All PNG paths resolve on disk.

Rank	Scene	Scene class	Lane M total HF	Lane A HF	One-line summary
10	electrophoresis_bench	instrument_heavy	24	8	Worst scene. 7 off_page + 1 region_overflow + 3 svg_label_overlap + 7 clipped_by_parent + 6 aspect_distorted. Tank overflows top by 171px; front_tools row overflows bottom; serological_pipette at 253% aspect distortion.
9	drug_dilution_workspace_dense	dense_clutter	19	4	4 off_page + 4 clipped_by_parent + 10 aspect_distorted. tube_rack_24 placeholder leak; tip_box aspect 45.65% wrong; tube_rack_2 at 45.31%.
8	crowded_bench_dense	dense_clutter	14	5	3 off_page + 2 region_overflow + 3 clipped_by_parent + 6 glassware aspect distortions at 5.16% each. front_tools region exceeds 260px budget by 34px.
7	drug_dilution_plate_workspace	composition	11	2	2 off_page + 2 clipped_by_parent + 6 aspect_distorted. tool_p200 and waste_container clip past viewport bottom; tip_box 45.65% aspect distortion.
6	staining_bench	composition	8	3	2 off_page + 1 region_overflow + 2 clipped_by_parent + 3 aspect_distorted. front_tools row clipped; primary staining_tray at 3.4% ratio.

Worst 5 thumbnails

#10 electrophoresis_bench (instrument_heavy, FAIL)

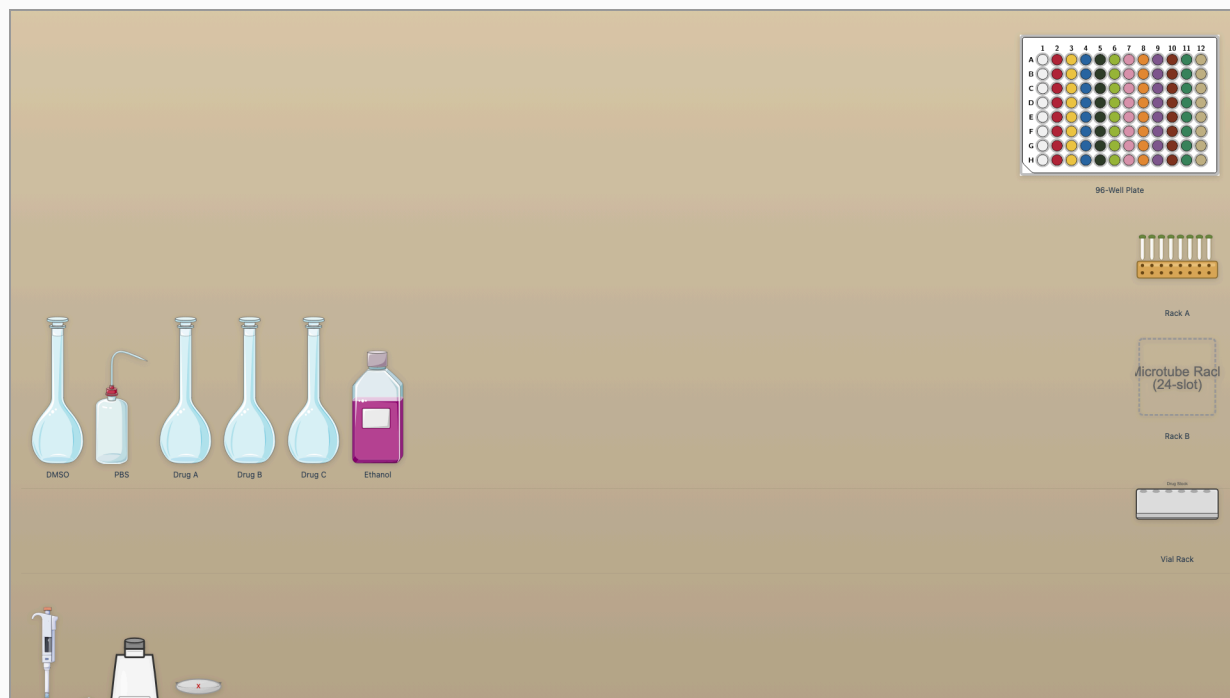
Lane A HF: 8 · Lane M total HF: 24



Worst scene. 7 off_page + 1 region_overflow + 3 svg_label_overlap + 7 clipped_by_parent + 6 aspect_distorted. Tank overflows top by 171px; front_tools row overflows bottom; serological_pipette at 253% aspect distortion.

#9 drug_dilution_workspace_dense (dense_clutter, FAIL)

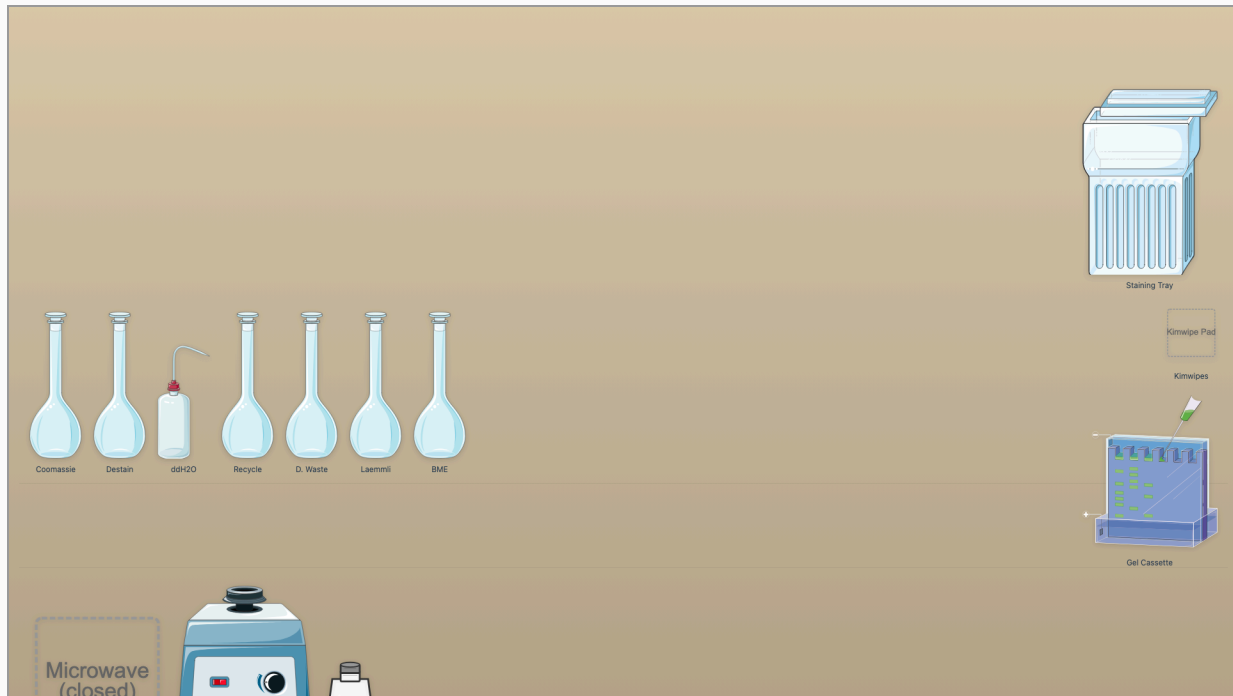
Lane A HF: 4 · Lane M total HF: 19



4 off_page + 4 clipped_by_parent + 10 aspect_distorted. tube_rack_24 placeholder leak; tip_box aspect 45.65% wrong; tube_rack_2 at 45.31%.

#8 crowded_bench_dense (dense_clutter, FAIL)

Lane A HF: 5 · Lane M total HF: 14



3 off_page + 2 region_overflow + 3 clipped_by_parent + 6 glassware aspect distortions at 5.16% each. front_tools region exceeds 260px budget by 34px.

#7 drug_dilution_plate_workspace (composition, FAIL)

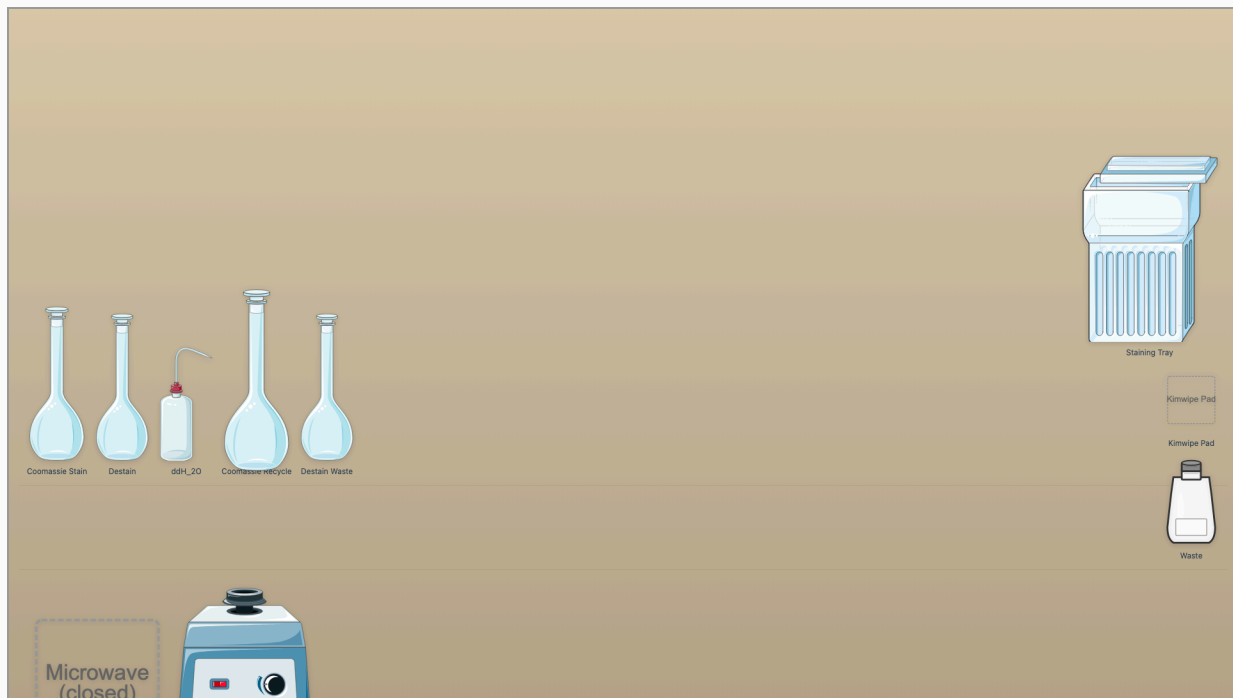
Lane A HF: 2 · Lane M total HF: 11



2 off_page + 2 clipped_by_parent + 6 aspect_distorted. tool_p200 and waste_container clip past viewport bottom; tip_box 45.65% aspect distortion.

#6 staining_bench (composition, FAIL)

Lane A HF: 3 · Lane M total HF: 8



2 off_page + 1 region_overflow + 2 clipped_by_parent + 3 aspect_distorted. front_tools row clipped; primary staining_tray at 3.4% ratio.

6. Diagnostic measurements (aggregate)

Numbers below come from `test-results/m0_static_summary/scorecard/scorecard.json`, `test-results/m0_static_summary/preccheck/visual_audit.json`, Lane M's extended count in `docs/active_plans/reports/m0_static_failed_scenes.md`, and Lane D's category rollup in `docs/active_plans/reports/m0_static_svg_completeness.md`.

Metric	Count	Source
Scenes processed	10	Lane A preccheck + Lane B scorecard
Verdicts FAIL	10	Lane B (8 are score 0 hard-fail FAIL; the 2 non-zero scores are PASS and MARGINAL per scorecard summary but the per-scene preccheck verdict is FAIL on all 10)
Aggregate layout score	152 / 1000	Lane B scorecard.md ranked table
Off-page placements	21	Lane A HARD_FAIL_TABLE totals
Region overflows	4	Lane A HARD_FAIL_TABLE totals
SVG-label overlaps	3 (all in electrophoresis_bench)	Lane A + Lane M cross-reference
Clipped artwork (Lane A column-set)	0	Lane A HARD_FAIL_TABLE totals
SVG-SVG overlaps	0	Lane A HARD_FAIL_TABLE totals
Label-label overlaps	0	Lane A HARD_FAIL_TABLE totals
<code>clipped_by_parent</code> HARD_FAIL (Lane M extended)	21	Lane M aggregate table
<code>aspect_distorted</code> HARD_FAIL (Lane M extended)	38	Lane M aggregate table
Placeholder asset references	2 (<code>microtube_rack_24_placeholder.svg</code> in <code>drug_dilution_plate_workspace</code> and <code>drug_dilution_workspace_dense</code>)	Lane A + Lane D
WRONG_ASSET references (Lane D)	14	Lane D aggregate category table

The aggregate score 152/1000 follows the scorer's contract: any scene with `hard_fails > 0` is floored to 0 regardless of weighted metric performance. Only `well_plate_96_zoom` (92) and `hood_basic` (60) clear the zero floor.

7. Exact commands used

All wall times are single-invocation, headless Chromium under Playwright.

Lane A precheck (9.180 s)

```
node experiments/css_native_layout/precheck.mjs \  
  'experiments/css_native_layout/templates/*.html' \  
  --out test-results/m0_static_summary/precheck \  
  --annotate on
```

Produced: `visual_audit.json`, `visual_audit.md`, `sizing_manifest.json`, `HARD_FAIL_TABLE.md`, and 10 PNGs under `test-results/m0_static_summary/precheck/`. The `--annotate on` flag warned that the optional Python annotation helper `_temp_annotate.py` is absent; core artifacts were still produced.

Lane B scorecard (under 100 ms)

```
node experiments/css_native_layout/score_layout.mjs
```

Reads `test-results/new0_css_native/audit` (the path is hardcoded in the script). Outputs were copied to `test-results/m0_static_summary/scorecard/scorecard.{json,md}` and the human-readable `SCORECARD_SUMMARY.md` was authored from `scorecard.json` in the same lane.

Lane D no-crop audit (standalone, no `src/` dependency)

```
node experiments/css_native_layout/no_crop_audit/inspect.mjs
```

Reported 16 cropping or distortion incidents across the same 10 templates. JSON output was copied verbatim to `test-results/m0_static_summary/no_crop_audit/no_crop_audit_results.json`. The SVG completeness rollup itself was assembled by a manual fallback that paired every `` reference with its enclosing `data-object-name` and categorized against the assets tree under `assets/equipment/`.

Lane K viewport sweep (17.9 s)

```
node test-results/m0_static_summary/viewport_sweep/_sweep.mjs
```

Renders each of the 10 templates at 1200x900, 1440x1000, and 1920x1080. Wrote 30 PNGs plus 30 JSON files (10 templates x 3 viewports = 60 artifacts) and `SWEEP_TABLE.md`.

8. Output paths

All paths are relative to repo root. Every listed file exists on disk at the time of this report; the PNGs were verified with `ls -la`.

- `test-results/m0_static_summary/precheck/visual_audit.json`
- `test-results/m0_static_summary/precheck/visual_audit.md`
- `test-results/m0_static_summary/precheck/sizing_manifest.json`
- `test-results/m0_static_summary/precheck/HARD_FAIL_TABLE.md`
- `test-results/m0_static_summary/precheck/bench_basic.png`
- `test-results/m0_static_summary/precheck/cell_counter_basic.png`
- `test-results/m0_static_summary/precheck/crowded_bench_dense.png`
- `test-results/m0_static_summary/precheck/drug_dilution_plate_workspace.png`
- `test-results/m0_static_summary/precheck/drug_dilution_workspace_dense.png`
- `test-results/m0_static_summary/precheck/electrophoresis_bench.png`
- `test-results/m0_static_summary/precheck/hood_basic.png`
- `test-results/m0_static_summary/precheck/microscope_basic.png`
- `test-results/m0_static_summary/precheck/staining_bench.png`
- `test-results/m0_static_summary/precheck/well_plate_96_zoom.png`
- `test-results/m0_static_summary/scorecard/scorecard.json`
- `test-results/m0_static_summary/scorecard/scorecard.md`
- `test-results/m0_static_summary/scorecard/SCORECARD_SUMMARY.md`
- `test-results/m0_static_summary/INDEX.html`
- `test-results/m0_static_summary/viewport_sweep/1200x900/` (10 PNG + 10 JSON)
- `test-results/m0_static_summary/viewport_sweep/1440x1000/` (10 PNG + 10 JSON)
- `test-results/m0_static_summary/viewport_sweep/1920x1080/` (10 PNG + 10 JSON)
- `test-results/m0_static_summary/viewport_sweep/SWEEP_TABLE.md`
- `test-results/m0_static_summary/viewport_sweep/_sweep.mjs`
- `test-results/m0_static_summary/no_crop_audit/no_crop_audit_results.json`
- `docs/active_plans/reports/m0_static_svg_completeness.md`
- `docs/active_plans/reports/m0_static_failed_scenes.md`

9. What cannot be tested because src/ is gone

Each bullet names a capability the upstream tests would normally cover but cannot exercise today because the runtime is absent.

- Runtime click dispatch. No `SceneRuntime` exists; no `compose_scene`, no event router, no click-target binding. A static PNG cannot prove click behavior.
- `ObjectStateChange` material updates. The protocol primitive that mutates `material_name`, `material_volume`, `held_material_name`, and `held_material_volume` cannot run.
- Validator preset hierarchical-target matching. Without a runtime, `correct_target`, `correct_choice`, `target_with_value`, `sequence_complete`, and `final_state_matches` cannot evaluate.
- Production 1920x1080 `well_plate_96_zoom` rect overflow. The runtime rect probe (`experiments/css_native_layout/well_plate_rect_probe/probe.mjs`) needs the built bundle; the static template-only run does not reproduce the runtime SVG rect coordinates for A1, E7, H12.
- Walker / Playwright spike runtime proofs. Visible-UI walkthroughs require the built app served from `dist/`; `dist/` is empty.
- Layout adapter recompute on state change. The CSS-native adapter spike (formerly `src/scene_runtime/layout/css_native_adapter.ts`) is gone; recompute behavior on object-state change is unobservable.
- Material setter primitives. The PROTOCOL_SPEC material setter family (`set_material_volume`, `transfer_material`, derived state flow) is not callable without `src/`.
- Cross-scene `SceneChange`. The protocol primitive that transitions scene context requires the runtime; only a single static template can be rendered at a time.

10. Next implementation target

The architecture extraction plan

`docs/active_plans/active/layout_manager_clean_start/architecture_extraction_plan.md` lays out milestones M0 through M6 for rebuilding the production layout manager under

`src/scene_runtime/layout/`. Lane E recommends M1 of that plan, scoped narrowly to two CSS-only fixes that do not need `src/` rebuilt and that produce the largest measurable static-evidence delta against the M0 baseline.

Lane M root cause 4.1 (`front_tools` region height) is the dominant multi-scene defect. Per Lane M's 5.1 "Easiest recoveries", a single CSS change to `front_tools` (reduce region height OR move tall tool cards out of `front_tools`) is forecast to recover approximately 17 `off_page` entries and approximately 21 `clipped_by_parent` HARD_FAIL entries across `bench_basic`, `crowded_bench_dense`, `drug_dilution_plate_workspace`, `drug_dilution_workspace_dense`, `electrophoresis_bench`, and `staining_bench`.

Lane M root cause 4.3 (aspect distortion is repo-wide) is the dominant single-class defect. Per Lane M's 5.2 "Per-asset card resizing", resizing pipette, tip_box, and glassware cards so each card aspect matches the natural SVG aspect is forecast to recover approximately 38 `aspect_distorted` HARD_FAIL entries across 9 of 10 scenes.

Both fix surfaces are pure CSS edits under `experiments/css_native_layout/styles/` and pure card-dimension changes inside the template HTML; neither requires `src/` to be repopulated. Both produce

measurable deltas on the next precheck + scorecard run.

Combined upper-bound recovery: up to 17 of 21 `off_page` hard fails, up to 21 of 21 `clipped_by_parent` extended hard fails, and up to 38 of 38 `aspect_distorted` extended hard fails. The 89 total Lane M hard fails could drop to a residual on the order of 10 to 15 (the electrophoresis tank top-clip at $y=-217$, the three center-cluster `svg_label_overlap` entries, the placeholder leak in two scenes, and the microscope wrong-asset). That residual maps cleanly onto the Lane M section 5.3 per-scene targeted fixes and the section 5.4 pedagogy review, neither of which is part of M1.

11. Files read

Top references consulted while writing this report:

- `test-results/m0_static_summary/scorecard/scorecard.md`
- `test-results/m0_static_summary/scorecard/SCORECARD_SUMMARY.md`
- `test-results/m0_static_summary/precheck/HARD_FAIL_TABLE.md`
- `test-results/m0_static_summary/viewport_sweep/SWEEP_TABLE.md`
- `docs/active_plans/reports/m0_static_failed_scenes.md`
- `docs/active_plans/reports/m0_static_svg_completeness.md`
- `docs/active_plans/active/layout_manager_clean_start/ONBOARDING.md`
- `docs/active_plans/active/layout_manager_clean_start/architecture_extraction_plan.md`
- `docs/REPO_STYLE.md`, `docs/MARKDOWN_STYLE.md`, `AGENTS.md`
- Filesystem listings of `test-results/m0_static_summary/precheck/*.png` and the subdirectories under `test-results/m0_static_summary/`.

Source Markdown: `docs/active_plans/reports/m0_static_visual_summary.md` · Thumbnails: `test-results/m0_static_summary/precheck/` · Lane F (HTML/PDF report) under `layout-manager-clean-start`.